

Matthew A. Carballo

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DATA ANALYST | DATA ENGINEER | DATA SCIENTIST

EDUCATION

University of North Carolina at Chapel Hill | Chapel Hill, NC Dec 2026

Bachelor's in Data Science, Sociology Concentration | Minor: Artificial Intelligence | GPA: 3.7/4.0

Relevant Courses: Data Collection & Management, Data Analysis & Visualization, Machine Learning, Applications of ML, Data & AI Ethics

Johnston Community College | Smithfield, NC May 2023

A.A.S. in Medical Assisting | GPA: 3.8/4.0 | Graduated with Honors | 200+ Clinical Hours

TECHNICAL SKILLS

Languages: Python, SQL, R

Data Engineering: PostgreSQL, Apache Airflow, dbt, Git

Analysis & ML: pandas, NumPy, scikit-learn, statsmodels, SHAP, Matplotlib, Seaborn, Jupyter, Excel

Methods: Data Cleaning, Visualization, Statistical Modeling, Machine Learning, NLP (LDA Topic Modeling), EHR Systems

PROJECTS

NYC 311 Service Requests: From Raw Data to Insights | Data Engineering Project May 2026

- Built an end-to-end **ELT pipeline** ingesting 3,000+ NYC 311 records daily from a public **REST API** into **PostgreSQL**, orchestrated with **Apache Airflow** for daily scheduling using **Python** and **SQL**
- Modeled data using **dbt** staging and mart layers to engineer calculated fields including resolution time and borough normalization; surfaced that Brooklyn leads all boroughs in complaints, Noise - Residential is the #1 complaint type, and Queens has the longest average resolution time

Hospital Readmission Risk Model | Data Science Project June 2026

- Built a **Random Forest classifier** on 100,000+ diabetic patient encounters to predict 30-day readmission (ROC-AUC 0.67); tested **SMOTE**, diagnosed that it distorted feature importance on mixed-type data, and switched to **balanced class weighting**, restoring clinically plausible predictor rankings
- Used **SHAP** values to identify prior inpatient visits and discharge disposition as the strongest readmission predictors and translated outputs into clinically interpretable insights

Readmission Risk MLOps Pipeline | ML Engineering Project Ongoing

- Productionizing a 30-day hospital readmission classifier into a deployable ML service, implementing **MLflow** experiment tracking and a **model registry** with staged promotion and rollback
- Developing a **FastAPI** endpoint for real-time risk scoring, containerized with **Docker** and deployed through **GitHub Actions CI/CD**, with **Evidently** drift monitoring and a **Streamlit** dashboard for **SHAP**-based explanations

Loan Outcome Prediction Model | Academic Project, STOR 320 July 2026

- Compared **Logistic Regression** and **Random Forest classifiers** on 45,000 loan applications; Random Forest achieved 92.95% accuracy and ROC-AUC of 0.975 versus 84.56% for logistic regression, with both models exceeding a 77.8% majority-class baseline
- Performed **feature engineering** including log-income transformation, one-hot encoding, and a **stratified train-test split**; identified loan-to-income ratio and interest rate as top predictors

Securing the AI Frontier: U.S.-New Zealand AI Policy Alignment | Academic Project, DATA 520 Ongoing

- Developing an **NLP pipeline** that applies **LDA topic modeling** to U.S. and New Zealand AI policy documents from the OECD AI Policy Observatory, with a validation framework using coherence scores, **Cohen's kappa**, **Jensen-Shannon divergence**, and **permutation testing**

EXPERIENCE

Handshake | AI Fellow | Remote Dec 2025 – Present

- Evaluated AI-assisted hiring tools through structured assessments, providing feedback on model outputs used in professional recruiting contexts and contributing to product iteration

UNC Urgent Care at Clayton | Medical Assistant Intern | Clayton, NC Jan 2023 – May 2023

- Recorded and analyzed patient data using EHR systems, improving data entry accuracy by 15% and streamlining daily reporting for clinical staff

LEADERSHIP

SHPE – UNC: Volunteered at a SolHacks hackathon, supporting registration and assisting attendees with coding challenges 2024 – Present

ALPFA – UNC: Supported chapter fundraising events and contributed ideas for future initiatives 2025 – Present